

# DHANUPRIYA A

Python / AI Developer - [LinkedIn](#) | [Portfolio](#)

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## PROFESSIONAL SUMMARY

**Python & AI developer with 4 years of experience** building scalable backend systems, enterprise AI applications, generative AI solutions, and real-time computer vision applications. Strong expertise in **Python, FastAPI, Django, Flask, asynchronous development, PostgreSQL, SQL Server, MySQL, and Snowflake**, with experience in **multi-tenant ERP/HRMS, payroll systems, REST APIs, and data pipelines**. Experienced in **LLM/AI integration, semantic search, text-to-SQL, document intelligence, computer vision, and AI-powered applications** using Gemini, OpenAI, YOLOv8, ArcFace, PyTorch, and ONNX Runtime. Skilled in **payroll engine development, vector search, ETL, PII detection, API security, GPU inference, testing, and Docker/Linux deployment**.

## CORE TECHNICAL SKILLS

|                                       |                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>                    | Python, SQL, Object-Orientated Programming, Async Programming                                                                                                                                |
| <b>Backend</b>                        | FastAPI, Django, Flask, REST APIs, API Design, Async Python, Uvicorn                                                                                                                         |
| <b>Generative AI / LLM</b>            | OpenAI GPT/GPT-4o, Google Gemini, MedGemma, LangChain, RAG, Text-to-SQL, Prompt Engineering, AI Model Integration                                                                            |
| <b>Machine Learning</b>               | Classification, Regression, Forecasting, Time Series, Anomaly Detection, Supervised Learning, Unsupervised Learning, Embeddings                                                              |
| <b>Computer Vision / AI Inference</b> | YOLOv8, OpenCV, MediaPipe FaceMesh, ArcFace, FAISS, LivePortrait, InsightFace, PyTorch, ONNX Runtime, GPU Inference                                                                          |
| <b>Data / Databases</b>               | PostgreSQL, SQL Server, MySQL, Oracle, MongoDB, Snowflake, ChromaDB, FAISS, SQLAlchemy, asyncpg, psycopg2                                                                                    |
| <b>Data Engineering</b>               | Apache Airflow, ETL, Parquet/JSON, JSON Flattening, Chunk-Based Processing, Metadata Processing, Azure Blob Storage                                                                          |
| <b>Enterprise Systems</b>             | Multi-Tenant Architecture, ERP/HRMS, Payroll Processing, Attendance-to-Payroll, Statutory Payroll, Salary Formula Evaluation, AST-Based Parsing, DAG Dependency Validation, Batch Processing |
| <b>Security</b>                       | JWT, RBAC, Argon2/bcrypt, Fernet Encryption, SSH Tunnelling, SQL Validation, PII Detection, Data Masking                                                                                     |
| <b>Cloud/AI Platforms</b>             | AWS S3, Google Vertex AI, Snowflake Cortex                                                                                                                                                   |
| <b>AI / Application Services</b>      | Gemini Image Generation, OpenAI GPT Image, GPT-4o Vision, Sarvam Speech-to-Text, ElevenLabs Text-to-Speech                                                                                   |
| <b>DevOps / Tools</b>                 | Docker, Jenkins, PM2, Linux, Git, GitHub/GitLab, pytest, Postman, Alembic, Pydantic, Paramiko, Streamlit, Jupyter, Google Colab, VS Code                                                     |

## WORK EXPERIENCE

### Software Engineer Backend developer—Inovatrik Technologies

Feb 2026 – Sep 2026 | Madurai, Tamil Nadu

- Architected and developed a **multi-tenant enterprise ERP/HRMS backend** using Python, FastAPI, asyncpg, and PostgreSQL, implementing tenant-isolated database routing, dynamic connection pooling, encrypted credentials, and Circuit Breaker-based resilience.
- Engineered a **high-performance payroll calculation engine** supporting gross-to-net processing, pro rata and daily/monthly wage calculations, overtime, shift incentives, deductions, loans, advances, arrears, and configurable salary components.
- Implemented **AST-based formula evaluation and DAG dependency validation** for safe configurable payroll formulas, preventing arbitrary code execution, self-references, and circular dependencies; automated **EPF, EPS, ESI, PT, and LWF** compliance.
- Integrated **biometric attendance with payroll** for LOP, overtime, shift allowances, and working-day calculations, with batch processing and optimised asynchronous database operations.
- Built **Smart Query**, a natural-language-to-SQL platform using FastAPI, Google Gemini, and ChromaDB, supporting semantic schema search, SQL generation/validation, read-only execution, database introspection, multi-tenancy, and secure database connectivity.
- Integrated and productionised **Gemini, OpenAI GPT/GPT-4o, MedGemma, LivePortrait, PyTorch, ONNX Runtime, and InsightFace** across AI image generation, AI avatar animation, prescription workflows, and enterprise database/data-masking solutions; supported deployment using Docker, Jenkins, Linux, and PM2.

### Python/AI Developer — Texila Educare Healthcare and Technology Enterprise Pvt.

May 2025 – Nov 2025 | Coimbatore, Tamil Nadu

- Led development of **NetricaAI**, a real-time CCTV monitoring and smart-attendance platform integrating face detection, face recognition, posture analysis, crowd detection, and automated attendance logging.
- Built the face-recognition pipeline using **YOLOv8, MediaPipe FaceMesh, ArcFace ONNX (512-D embeddings), and FAISS/cosine similarity**, with webcam/file-based employee onboarding and SQL Server integration.
- Engineered resilient **FFmpeg-based RTSP video ingestion** with auto-reconnect, frame skipping, queue buffering, multithreaded processing, per-camera diagnostics, and FPS monitoring for multiple live streams.
- Developed **Flask/FastAPI backend services** for camera control, ROI configuration, MJPEG streaming, attendance, crowd alerts, authentication, and event validation, including entry/exit attendance logic and fallback mechanisms.
- Implemented **ROI-based crowd detection and DFS-based proximity analysis**, with posture metadata and snapshot capture; optimised inference using CUDA, ONNX Runtime, caching, and asynchronous background processing.
- Designed SQL Server schemas and optimised read/write operations, logging, monitoring, and exception handling; prepared the application for **Docker-based deployment** and collaborated with IT/DevOps on production readiness.

### Python/ML Developer—PBS Info Systems Pvt Ltd

Jan 2024 – Apr 2025 | Madurai, Tamil Nadu

- Developed **enterprise AI and data solutions** using Python, Snowflake, generative AI, and cloud-based data pipelines for business intelligence and automation.
- Built a real-time **Document AI ETL pipeline** using Snowflake for automated vendor document extraction, metadata-driven validation, and operational monitoring through a Streamlit dashboard.
- Developed **hybrid semantic search** using Snowflake Cortex Search, Arctic Embed, and LangChain, and implemented LLM-powered **Text-to-SQL with Cortex Analyst** for natural-language enterprise analytics.
- Developed a **Cortex Agent platform using FastAPI and React** with YAML-driven schema routing, JWT authentication, interactive data exploration, and real-time visualisation.

- Built **forecasting and anomaly-detection pipelines** for time-series and financial transaction data using supervised/unsupervised approaches.
- Designed high-throughput Python **ETL pipelines using Apache Airflow and Azure Blob Storage** for Parquet/JSON datasets, optimising memory usage through chunk-based processing and nested JSON flattening.

### Junior Python Developer—Dot Com Infoway Ltd.

Sep 2022 – Dec 2023 | Madurai, Tamil Nadu

- Developed an **AI-powered chatbot for women's safety and health** using Django, NLP, OpenAI LLM services, voice-to-text accessibility, and keyword/topic filtering.
- Developed an **image upscaling platform** using OpenCV, NumPy, and EDSR deep-learning models to enhance low-resolution images up to 8x while preserving image detail.
- Diagnosed and resolved **SMTP configuration issues** for the Akshaya India website, restoring outgoing email delivery and supporting audit requirements.
- Developed and maintained **production-ready Python applications**, focusing on scalability, maintainability, documentation, testing, and reliable deployment practices.
- Collaborated with **cross-functional teams** to ensure applications met functional, performance, and compliance requirements.

### EDUCATION

- Master of Computer Applications (MCA) — The American College | 2019 – 2022 | 79%
- Bachelor of Computer Applications (BCA) — E.M.G. Yadava Women's College | 2016 – 2019 | 79%
- Higher Secondary Education (12th) — Nirmala Girls Higher Secondary School | 2016 | 83%
- Secondary Education (10th) — Nirmala Girls Higher Secondary School | 2014 | 91%

### ACHIEVEMENTS

- **Appreciated by Texila Educare Healthcare and Technology Enterprises** for delivering a remarkable contribution by developing and preparing the deployment of a **NetricaAI CCTV surveillance** project, recognised for teamwork, problem-solving, and going the extra mile (2025).
- Recognised for successfully implementing Enterprise Search at **PBS Info Systems**; published Transforming Enterprise Search with **AI Using Snowflake Cortex Search on Medium**. (2024)
- Awarded "New Technology Enthusiast" (2023) by **DotComInfoway**.
- Recipient of the **Academic Proficiency Certificate of Merit Award** at E. M. G. Yadava Women's College (2019).